

RESEARCH

Open Access



Biochar enhances methane uptake in engineered green roof substrate

Imrul Kayes^{1,2*} , Md Abdul Halim^{1,3} , Wenxi Liao⁴ , Md Rezaul Karim¹ , Melanie A. Sifton¹  and Sean C. Thomas^{1*} 

Abstract

Biochar is increasingly applied to soils to enhance carbon sequestration and mitigate greenhouse gas emissions, yet its role in regulating methane (CH₄), a potent greenhouse gas, in engineered substrates remains poorly understood. In particular, there is a lack of field data on how biochar amendments influence CH₄ fluxes from engineered substrates in urban environments, where substrate composition, shallow profiles, and intensive management fundamentally differ from natural soils. Here, we present a five-year period (2020–2024) field study examining the effects of biochar amendment on CH₄ fluxes from engineered substrates used in extensive green roofs dominated by stone-crop species (*Phedimus kamtschaticus* and *Sedum* spp.). Biochar-amended modules (~5% v/v; 20 t ha⁻¹) consistently exhibited greater CH₄ uptake than unamended controls across seasons, with peak uptake rates in spring 2023 nearly fivefold higher (-1.91 ± 0.25 vs. -0.40 ± 0.10 nmol·m⁻²·s⁻¹). In addition, biochar addition was not accompanied by increased CO₂ emissions. Variability in CH₄ uptake was strongly associated with substrate moisture and water vapour flux, suggesting that biochar modifies gas diffusivity and oxygen availability through moisture-mediated physical controls that favour microbial CH₄ oxidation. These results demonstrate that biochar can substantially enhance CH₄ uptake in engineered green roof substrates and extend the application of biochar-based mitigation strategies to engineered systems beyond conventional soils. The mechanistic insights provided here are broadly relevant to understanding CH₄ cycling in biochar-amended engineered substrates.

Highlights

- Biochar turns green roof substrates into consistent methane-absorbing systems.
- Methane uptake increased without increasing carbon dioxide emissions.
- Green roofs can be engineered as passive systems for greenhouse gas mitigation.

Keywords Biochar, Methane flux, Engineered substrates, Green roof substrates, Greenhouse gas fluxes, Urban soils

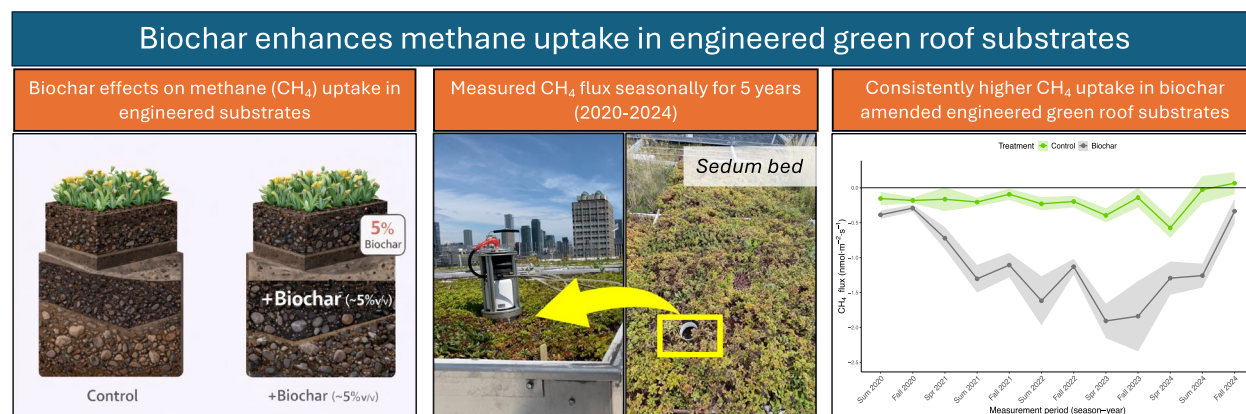
*Correspondence:

Imrul Kayes
imrul.kayes@mail.utoronto.ca
Sean C. Thomas
sc.thomas@utoronto.ca

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Graphical Abstract



1 Introduction

Methane (CH_4) is a potent greenhouse gas with a high near-term warming potential, and small changes in its surface–atmosphere exchange can exert disproportionate impacts on climate forcing (Saunois et al. 2020; Mar et al. 2022). Biochar, a carbon-rich material produced from biomass pyrolysis, has been widely studied for its ability to enhance carbon sequestration and modulate greenhouse gas fluxes through persistent alterations to substrate physical structure, chemical properties, and microbial habitat (Lehmann and Joseph 2009; Spokas 2010). However, biochar effects on CH_4 cycling remain uncertain in many ways, particularly regarding the relative roles of substrate moisture dynamics, gas diffusivity, redox conditions, and microbial activity in regulating CH_4 production and oxidation (Le Mer and Roger 2001; Conrad 2007; Wu et al. 2019). Although biochar-induced changes in CH_4 flux have been widely studied in agricultural soils, several key uncertainties remain for engineered substrates. First, most evidence derives from short-term experiments, providing little insight into the persistence of biochar effects through time. Second, the mechanisms linking biochar, substrate moisture dynamics, gas diffusivity, and CH_4 uptake remain poorly resolved under field conditions. Third, virtually no studies have quantified biochar effects on CH_4 fluxes from extensive green roof substrates, which differ fundamentally from mineral soils in depth, organic matter content, and hydrological behavior. Engineered substrates, characterized by shallow profiles, high organic matter content, and strong constraints on aeration and water dynamics, differ fundamentally from mineral soils and may therefore exhibit distinct biochar– CH_4 interactions. Resolving how biochar influences CH_4 flux in these systems is essential

for extending biochar-based climate mitigation strategies beyond conventional soil environments widely available across urban green infrastructure.

Green roofs represent a rapidly expanding class of engineered substrates within urban green infrastructure and are designed primarily to retain stormwater, moderate building energy use, and support vegetation under strict structural load limits (Oberndorfer et al. 2007; Wooster et al. 2022; Mihalakakou et al. 2023). The potential climate benefits of green roofs through carbon sequestration are also significant, as both vegetation and the organic carbon embedded in green roof substrates are increasingly recognized as important carbon sinks within urban green infrastructure systems (Whittinghill et al. 2014; Shafique et al. 2020). From a biogeochemical perspective, green roofs function as shallow, organic-rich engineered systems in which moisture dynamics, aeration, and microbial activity are tightly coupled, making them especially sensitive to substrate amendments that modify physical structure and gas transport.

The capacity of green roofs to exchange non- CO_2 GHGs, particularly CH_4 , has only recently received attention (Halim et al. 2022). While several studies have assessed carbon sequestration and CO_2 fluxes, substrate-driven GHG exchange remains a critical and underexplored component of green roof climate performance. Evidence is especially limited for extensive green roofs dominated by *Sedum* or *Phedimus* mats, where only a handful of studies have quantified CH_4 fluxes directly. High organic matter content—often incorporated to enhance water retention and plant performance (Hill et al. 2017; Xue and Farrell 2020)—may also promote CH_4 production under localized anaerobic conditions, with reported fluxes comparable to those from organic

mulches (Kayes et al. 2025). This highlights the need to quantify CH_4 exchange and identify mechanisms regulating methane cycling in these engineered systems.

Biogenic CH_4 flux reflects the balance between methanogenesis in anoxic or hypoxic microsites and methanotrophy in oxic zones, with both processes strongly influenced by substrate physical and chemical properties (Le Mer and Roger 2001; Conrad 2007). Moisture and temperature are dominant controls, with CH_4 oxidation typically peaking at intermediate substrate moisture content and declining under excessively wet or dry conditions due to oxygen limitation or microbial water stress (Castro et al. 1995; Borken et al. 2006). In green roofs, moisture and temperature have likewise been identified as the dominant physical drivers regulating CH_4 uptake (Teemusk et al. 2019; Halim et al. 2022). Soil pH is also an important chemical control; while earlier studies suggested optimal methanotrophic activity near neutrality, ranging between 6.6–7.5. (Krulwich et al. 2007), subsequent work has shown active CH_4 oxidation under both acidic and alkaline conditions (Trotsenko and Khmeleina 2002; Yao et al. 2023). Soil texture and structure further modulate CH_4 exchange by controlling gas diffusivity through air-filled porosity, while surface evaporation can indirectly enhance oxidation by promoting aeration in near-surface layers (Lehmann et al. 2018; Smith et al. 2018; Abeyasinghe et al. 2022). Because methanotrophs convert CH_4 to CO_2 , CH_4 and CO_2 fluxes may covary through both stoichiometric coupling and shared environmental drivers (Zheng et al. 2018; Ding et al. 2024). These controls are inherently seasonal, emphasizing the importance of long-term field observations to resolve treatment effects under variable climatic conditions (Wagner-Riddle et al. 2017).

Green roof substrates are engineered mixtures of lightweight organic matter and porous mineral aggregates designed to balance drainage, water retention, and load constraints (A'saf et al. 2020). While these materials support drainage and plant survival, over time, these substrates are prone to compaction and structural degradation, reducing porosity and altering moisture regimes (De-Ville et al. 2018; Yang and Davidson 2021). Rooftop exposure further enhances evaporative demand, often necessitating irrigation, which has been shown to increase CH_4 uptake by alleviating moisture limitation (Halim et al. 2022). While green roof substrates generally emit CO_2 , they typically function as weak CH_4 sinks, and practical strategies to strengthen this sink capacity remain largely unexplored.

Biochar has been shown to enhance CH_4 uptake across a range of agricultural, forest, and urban soils (Wu et al. 2019; Nan et al. 2020; Li et al. 2024; Kayes

et al. 2025), largely through improvements in porosity, aeration, and microbial habitat (Lehmann and Joseph 2009). In green roof systems, biochar has been reported to improve plant performance, stormwater retention, and nutrient availability (Chen et al. 2018; Liao et al. 2023; Lee and Kwon 2024), while reducing substrate wind erosion when applied as granulated biochar (Liao et al. 2022). Additionally, biochar can create a stable and favorable environment for methane-oxidizing microbes by altering substrate bulk density, electrical conductivity, and pH. Despite these benefits, its influence on CH_4 fluxes from green roof substrates has not been evaluated. Although biochar is highly recalcitrant, with carbon residence times on the order of centuries (Spokas 2010), transient increases in CO_2 efflux from labile carbon fractions have been reported (Wang et al. 2016), underscoring the need for long-term field assessment in engineered systems.

Urban areas now host more than half of the global population (United Nations, 2025), yet their contribution to global biogeochemical cycles remains poorly constrained. While natural soils are recognized as CH_4 sinks, urban ecosystems are often overlooked. Green roofs are widely promoted for stormwater retention, energy savings, and biodiversity, but their role in regulating GHG fluxes is rarely examined. Biochar is known to enhance soil carbon sequestration and stimulate CH_4 uptake in terrestrial systems, but evidence for its function in green infrastructure remains scarce. A recent meta-analysis found that biochar generally improves ecosystem functions across green infrastructure types, including reduced CO_2 and N_2O emissions, though CH_4 responses were variable (Liao et al. 2023). However, prior studies directly testing biochar–GHG interactions have been limited to constructed wetlands, where elevated CH_4 emissions are common (Guo et al. 2020; Chen et al. 2020). No field study has directly tested whether biochar enhances atmospheric CH_4 uptake in extensive green roof substrates, nor whether any observed effects persist across multiple growing seasons.

In this study, we quantified CH_4 , CO_2 , and H_2O fluxes from biochar-amended and control extensive green roof systems over intra- and inter-seasonal time-scales during a five-year period. We hypothesized that biochar amendment would enhance CH_4 uptake by modifying substrate physicochemical properties, with effects mediated by changes in moisture dynamics, evaporation, and CO_2 flux. We further expected that while the direction of biochar effects would remain consistent over time, their magnitude would vary seasonally.

2 Methods

2.1 Green roof design and substrates

The experiment was installed in fall 2019 at the Green Roof Innovation Testing Laboratory (GRIT Lab II) at 1 Spadina Crescent at University of Toronto's downtown St. George Campus, in Toronto, Ontario, Canada (Fig. S1a). GRIT Lab II's green roof modules were sized 1.83 m×1.83 m, with 15 cm substrate depth and slope of 2°. While GRIT Lab II consists of a total of 52 roof modules, the focus of this study was only on 24 green roof modules cultivated over the course of the trial using pre-grown stonecrop mats. The stonecrop mats were dominated (>95%) by orange stonecrop [*Phedimus kamtschaticus* (Fisch. & C.A.Mey.) 't Hart], with small amounts of white stonecrop [*Sedum album* L.], and tasteless stonecrop [*Sedum sexangulare* L.].

The experimental green roof media was a lightweight high-organic green roof mix ("Eco-blend" supplied by Bioroof Systems Inc., Burlington, Canada) with an organic matter content of 75 ± 2% (mean ± SE) (Liao et al. 2024). The biochar amendment was created using sugar maple [*Acer saccharum* L.] sawdust pyrolyzed at ~700 °C, with 10–15 min of residence time (Haliburton Forest and Wildlife Reserve, Canada). Biochar and substrate properties are presented in supplementary Table S1.

2.2 Experimental design

Green roof modules were arranged in a completely randomized design with one factor (control vs. biochar) and 12 replicates per level ($n=24$). Two non-vegetated bare-substrate modules of identical dimensions were also included for reference. CH₄, CO₂ and H₂O flux measurements were conducted from summer (June–August) 2020 through fall (September–November) 2024, spanning five growing seasons. Measurements for spring (March–May) 2022 and summer 2023 were not collected due to logistical constraints and were therefore excluded from the formal analysis. Biochar was incorporated into each treated green roof module at a rate of 20 t ha⁻¹ (equivalent to ~5% v/v), an application level previously identified as optimal for enhancing plant performance (Gale and Thomas 2019). Each green roof module received uniform irrigation of 20–30 L of municipal water twice weekly during drought periods. This irrigation regime was designed primarily as part of operational maintenance to sustain plant survival during dry periods and generally did not generate any surface runoff from the modules. Additional details on the experimental modules and their hydrological performance are given by Saade et al. (2023).

The full experimental design included 24 vegetated green roof modules measured across 12 seasonal measurement campaigns, yielding a maximum possible

sample size of 288 module-level observations. Following quality control, nine observations were excluded due to collar displacement during measurement campaigns, resulting in a final dataset of 279 observations used for all gas flux analyses. This sampling structure underlies the denominator degrees of freedom reported in the mixed-effects model analyses.

2.3 Data collection

In each replicate roof module, one PVC collar (10 cm diameter, 10 cm height) was inserted into the substrate in early May 2020, prior to initiating seasonal gas flux measurements conducted from summer 2020 through fall 2024. The collars were inserted 3–5 cm deep into the substrate to prevent leakage, following the method of Jovani-Sancho et al. (2018), and placed at least 30 cm away from the module edge to avoid interference from plant roots. Any aboveground plant material rooted within the collar was trimmed off at the time of installation and subsequent gas flux measurements to eliminate aboveground plant-mediated gas fluxes.

For gas-exchange measurements, we used a Los Gatos Ultraportable Greenhouse Gas Analyzer (UGGA: Los Gatos Research, San Jose, CA, USA), with measurements based on off-axis integrated cavity output spectroscopy. This was connected to a pressure-equilibrated chamber (LI-8100–102 opaque survey chamber, volume 854.2 cm³; LI-COR Environmental, Lincoln, NE, USA) in a closed dynamic system with a nominal flow rate of 0.5 L/min. This setup provided real-time measurements of CH₄, CO₂, and water vapor (H₂O), with respective accuracies of 2, 300, and 100 ppb.

Substrate temperature was recorded adjacent to each sampling location at a depth of ~5 cm using a Flip Thermometer (PC, Toronto, Canada; Model 21,010,083) with a precision of 0.5 °C. The substrate moisture content was measured at a depth of ~5 cm using a CS-SM2 soil moisture sensor (accuracy: ±2.5%) (CredoSense Inc., Toronto, Canada). Measurements were scheduled to avoid precipitation and irrigation in the preceding 48 h, mitigating transient gas flux effects related to the "Birch effect" (Birch 1958; Schurman and Thomas 2021). Each measurement session lasted approximately three minutes, with data logged at one-second intervals. A subset of substrates was sampled during every gas measurement event to determine pH and EC (electrical conductivity) in the laboratory. The pH and EC were determined in a 1:2 suspension of substrates in deionized water at room temperature using a pH/mV Meter (IQ Scientific Instruments, USA) and an Orion Star A112 benchtop conductivity meter (Thermo Scientific, USA), respectively. In each case, triplicate measures were conducted.

2.4 Flux calculations

Output files from the LGR gas analyzer were post-processed in R version 4.3.2 (R Core Team 2024) with an algorithm to estimate flux rates (dC/dt) of concentrations of CH_4 , CO_2 and H_2O (Halim et al. 2022, 2024). First, a “dead band” of initial 30 s was eliminated from each measurement, which is affected by artifacts triggered by the closing of the chamber (Hoffmann et al. 2017; Halim et al. 2024). After this exclusion, we applied a straightforward algorithm using the Pearson correlation coefficient (r) between concentration and time, focusing specifically on CO_2 concentration data to identify the optimal time window (50–60 s) for flux calculations. The time window yielding the highest r value for CO_2 was selected for both CO_2 and CH_4 gases in subsequent flux calculations. We selected a CO_2 -based time window for CH_4 because CO_2 fluxes tend to exhibit less noise than CH_4 , allowing potential leakage or pressure artifacts to be more easily detected from CO_2 data than from CH_4 . Figure S1 shows an example slope calculation from the raw data stream following the algorithm described above.

To determine dC/dt for calculating CO_2 and CH_4 fluxes, we used either linear or non-linear regression (Hüppi et al. 2018) depending on the statistical properties of the concentration data. If the quadratic coefficient of the concentration data was not statistically significant ($p > 0.05$), we used linear regression to determine dC/dt and calculated the flux as Li-COR (2015) (Li-Cor 2015):

$$F = \frac{10VP_0}{RS(T_0 + 273.15)} \frac{dC}{dt} \quad (1)$$

where F is the flux of water-corrected CH_4 , CO_2 or H_2O , V is total volume (sum of chamber head-space volume and the volume inside the collar aboveground) (cm^3), P_0 is the initial pressure (kPa), R is the Universal Gas Constant ($0.83144598 \text{ m}^3\text{kPa}\text{K}^{-1}\text{mol}^{-1}$), S is the soil surface area (cm^2), T_0 is the initial air temperature ($^\circ\text{C}$), and dC/dt is the initial rate of change in the water-corrected CH_4 , CO_2 or H_2O mole fraction. If the quadratic coefficient was significant ($p < 0.05$), we chose the non-linear regression approach. Throughout this paper, we expressed CH_4 fluxes as $\text{nmol m}^{-2}\text{s}^{-1}$ and, CO_2 and H_2O fluxes as $\mu\text{mol m}^{-2}\text{s}^{-1}$. We fitted the following empirical equation (Welles et al. 2001) to the data points of the selected time window:

$$C'_{(t)} = C'_x + (C'_0 - C'_x)e^{-a(t-t_0)} \quad (2)$$

where $C'_{(t)}$ is the instantaneous water-corrected mole fractions for the gases, C'_0 is the value of $C'_{(t)}$ when the chamber just closed, C'_x is the asymptote parameter, a specifies the curvature of the fit, and t_0 is time (s) when the chamber closed. The parameters a , t_0 , C'_x , and C'_0 were

estimated from the fitted nonlinear regression. Subsequently, the following equation (Eq. (3)), derived from Eq. (2) (at $t=t_0$), was used to calculate the initial rate of change of the water-corrected $\text{CH}_4/\text{CO}_2/\text{H}_2\text{O}$ mole fraction (Li-Cor 2015):

$$\frac{dC}{dt} = a(C'_x - C'_0) \quad (3)$$

The value of dC/dt obtained from Eq. (3) was used in Eq. (1) to calculate the CH_4 , CO_2 and H_2O flux. Overall, using the above algorithm, 18% of the total CH_4 flux, 16% of the total CO_2 flux, and 9% of H_2O flux were calculated using the nonlinear regression approach (primarily for high flux rates). The average of the three replicated flux measurements of each gas, taken at each collar per campaign, was used for further analyses.

2.5 Statistical analysis

Statistical analyses were conducted to evaluate the effects of biochar treatment and measurement period (season $\times$ year) on CH_4 , CO_2 , and H_2O fluxes across the study years. To evaluate these effects, a linear mixed-effects model (LMMs) were applied using the “lmer” function from the *lme4* package in R (Bates et al. 2015). Fixed effects included biochar treatment, measurement period (season-year) and their interaction (treatment $\times$ measurement period (season-year)). To account for repeated measurements, Collar was included as a random intercept, while measurement period nested within year was included to capture temporal dependence across measurement years. Residual diagnostics were inspected to verify normality and homoscedasticity assumptions.

The significance of fixed effects was evaluated using Type III Analysis of Variance (ANOVA) with Satterthwaite’s method for estimating denominator degrees of freedom, implemented through the “Anova” function in the “lmerTest” package (Fox et al. 2001). To further explore significant fixed effects and interactions, pairwise post hoc comparisons were performed using estimated marginal means (EMMs) obtained from the fitted linear mixed-effects models with the “emmeans” package in R. Treatment contrasts were evaluated within each season, and Sidak adjustment was applied to control for multiple comparisons ($\alpha = 0.05$). This model-based approach retains the random-effects structure, providing statistically robust and interpretable pairwise inferences. Non-vegetated bare-substrate modules were excluded from the formal statistical analysis due to the limited number of replicates, which precluded adequate statistical power; their results are therefore presented descriptively only.

2.6 Structural equation modeling

We used structural equation modeling (SEM) to evaluate direct and indirect associations between biochar addition, substrate properties, and gas and water-vapour fluxes. Because measurements were repeated within collars and across seasons, we implemented a *piecewiseSEM* in R (v4.4.0) using component regressions fit with linear mixed-effects models (*lme4*), with random intercepts for collar and season to account for non-independence (Gunzler et al. 2013; Danner et al. 2015; Lefcheck 2016). Treatment was coded as an exogenous binary predictor (0=Control, 1=Biochar). Substrate moisture, temperature, EC, and pH were specified as mediators; H₂O, CO₂, and CH₄ fluxes were specified as endogenous responses.

The a priori model specified that biochar addition could influence substrate moisture, temperature, EC, and pH, and that these substrate properties could in turn influence H₂O, CO₂, and CH₄ fluxes. We assumed no a priori causal direction among fluxes of CH₄, CO₂, and H₂O; therefore, we did not specify directed paths among the fluxes. Instead, we modeled their shared, potentially unmeasured drivers by allowing residual covariances among fluxes within the piecewise SEM framework. All component models were fit as linear mixed-effects regressions with random intercepts for collar and season; when this crossed random-effects structure resulted in singular fits, we simplified the random structure to obtain stable model fits. Overall model adequacy was evaluated using Fisher's C statistic, where non-significant values ($p > 0.05$) indicate consistency between the hypothesized network and the observed covariance structure. Standardized path coefficients, marginal R² values, and the SEM diagram were used to interpret the final model.

3 Results

3.1 Biochar amendment strengthens CH₄ uptake across seasons

Linear mixed-effects models (LMMs) showed that biochar treatment had a highly significant effect on CH₄ flux ($F_{1,255} = 136.52$, $p < 0.001$) in green roof substrates. The main effect of season was not significant ($F_{11,255} = 0.59$, $p = 0.84$); however, the interaction between treatment and season was significant ($F_{11,255} = 3.67$, $p < 0.001$), indicating that the effect of treatment on CH₄ flux varied across seasons (Table 1).

CH₄ fluxes from green roof substrates showed clear treatment variations across measurement campaigns over the years (Fig. 1; Table S3). During the initial measurement years (2020–2021), CH₄ uptake was modest in both treatments, but by summer 2021 biochar-amended modules exhibited significantly stronger uptake than those without biochar ($p < 0.001$). From 2022 onward,

Table 1 Gas fluxes (CH₄, CO₂ and H₂O) from substrates under pre-grown stonecrop-mat extensive green roof modules in response to biochar treatment and season. Significant p -values are shown in bold

Greenhouse gas	Effects	Mean sq	Df	F-value	p -value
CH ₄	Treatment	57.3	1	136.52	< 0.001
	Season	0.246	11	0.587	0.838
	Treatment × Season	1.54	11	3.671	< 0.001
CO ₂	Treatment	2.648	1	0.316	0.579
	Season	315.48	11	37.66	< 0.001
	Treatment × Season	16.694	11	3.671	< 0.029
H ₂ O	Treatment	2,483,904	1	28.99	< 0.001
	Season	1,080,987	11	12.620	< 0.001
	Treatment × Season	310,866	11	3.629	< 0.001

biochar consistently enhanced CH₄ uptake relative to the no-biochar treatment across all seasons (all $p < 0.05$). Among individual measurement campaigns, the largest numerical treatment difference was observed in spring 2023, when CH₄ uptake in biochar-amended green roof modules reached -1.91 ± 0.25 nmol·m⁻²·s⁻¹ compared with -0.40 ± 0.10 nmol·m⁻²·s⁻¹ in modules without biochar ($t = -5.84$, $p < 0.001$). Significant treatment contrasts also occurred in fall 2021 ($p = 0.02$), summer 2022 ($p = 0.001$), and spring 2024 ($p = 0.007$), confirming the persistence of enhanced CH₄ uptake in biochar-treated substrates. Across the five-year period, CH₄ fluxes exhibited strong temporal structure, with uptake peaks in spring and fall and weaker uptake in mid-summer, reflecting concurrent variation in substrate moisture and temperature (supplementary Fig. S2). By 2024, although CH₄ uptake had slightly weakened relative to 2023, biochar-treated modules still maintained higher uptake rates (-1.26 ± 0.17 nmol·m⁻²·s⁻¹) than the no-biochar controls (-0.03 ± 0.19 nmol·m⁻²·s⁻¹; $p < 0.001$) across all seasons (Fig. 1; Table S3). Bare substrate fluxes were also measured but excluded from formal statistical analyses due to lower replication; their descriptive trends, showing minimal CH₄, CO₂, and H₂O exchange relative to vegetated modules, are summarized in supplementary Tables S3–S5.

3.2 CO₂ flux varies seasonally with limited biochar effect

Results from LMMs showed that biochar treatment had no significant effect on CO₂ flux ($F_{1,255} = 0.32$, $p = 0.58$). In contrast, the main effect of season was highly significant ($F_{11,255} = 37.66$, $p < 0.001$), indicating substantial temporal variation in CO₂ flux. The interaction between treatment and season was also significant ($F_{11,255} = 1.99$,

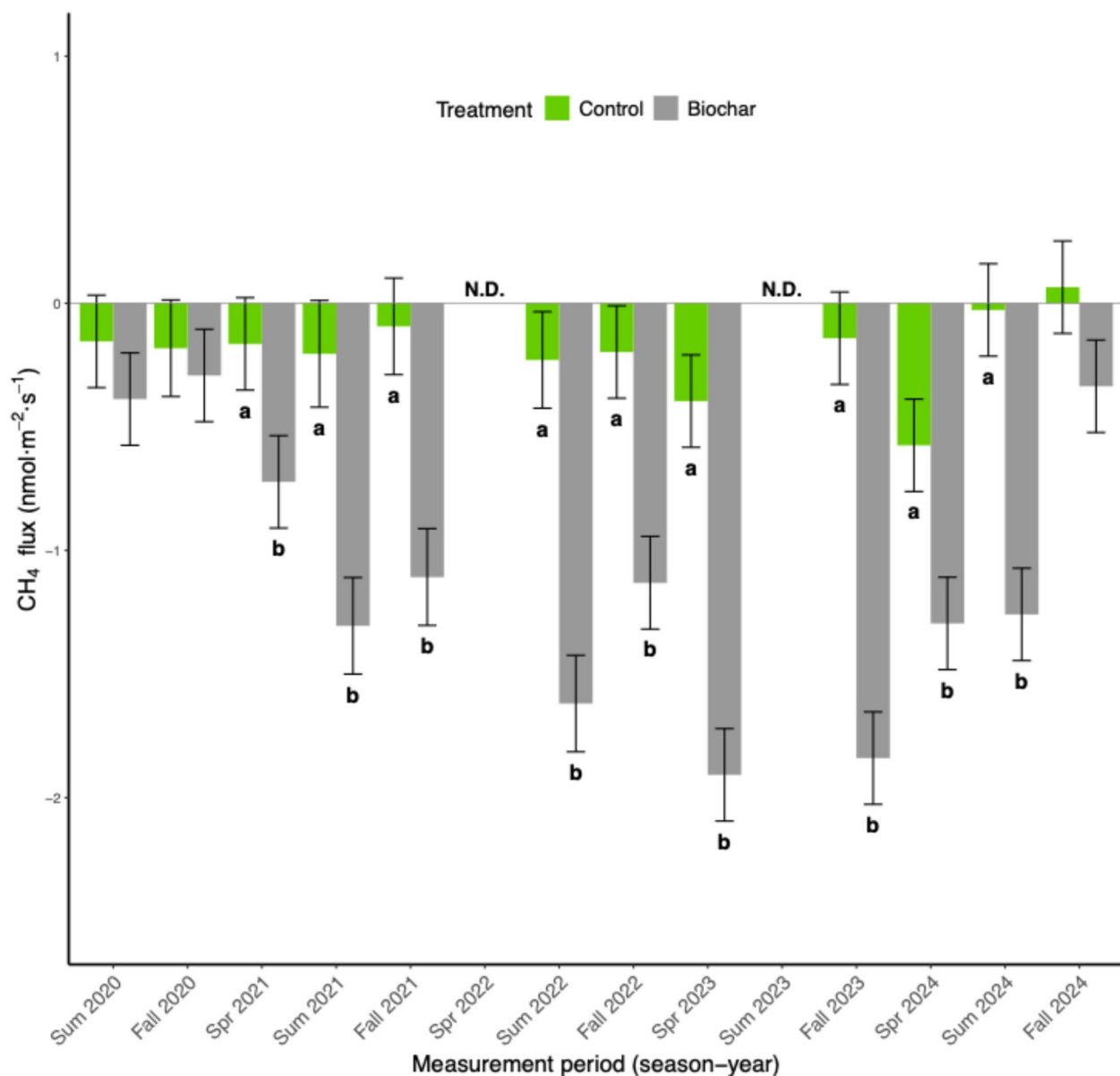


Fig. 1 Seasonal variation in mean ($\pm$ SE) CH₄ flux from stonecrop-vegetated extensive green roof modules with and without biochar amendment from summer 2020 to fall 2024. Bars represent estimated marginal means derived from a linear mixed-effects model. Within each measurement period (season-year), different letters indicate significant differences between biochar and control treatments based on Sidak-adjusted pairwise comparisons ($p < 0.05$). Negative flux values indicate net atmospheric CH₄ uptake by the substrate. N.D. denotes periods for which measurements were not conducted (spring 2022 and summer 2023)

$p = 0.03$), suggesting that treatment effects on CO₂ flux varied modestly across seasons (Table 1).

CO₂ flux showed significant temporal variation, with moderate emissions during early establishment (2020–2021) and pronounced temporal fluctuations thereafter (Fig. 2a; Table S4). Biochar treatment did not exert a consistent effect on CO₂ flux ($p > 0.05$ for most comparisons), though a significant treatment $\times$ season interaction was evident (Fig. S3). In fall 2022, modules without biochar

exhibited the highest CO₂ emissions ($15.7 \pm 1.2 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$), significantly exceeding those from biochar-amended modules ($11.4 \pm 1.6 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; $p = 0.01$). By spring 2023, both treatments showed comparable fluxes ($\sim 9\text{--}10 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; $p > 0.3$), and this pattern persisted through 2024, with modest seasonal peaks in spring and fall (Fig. 2a; Table S4). Overall, CO₂ fluxes followed a strong seasonal pattern, peaking in spring and fall and declining in summer, with interannual variability.

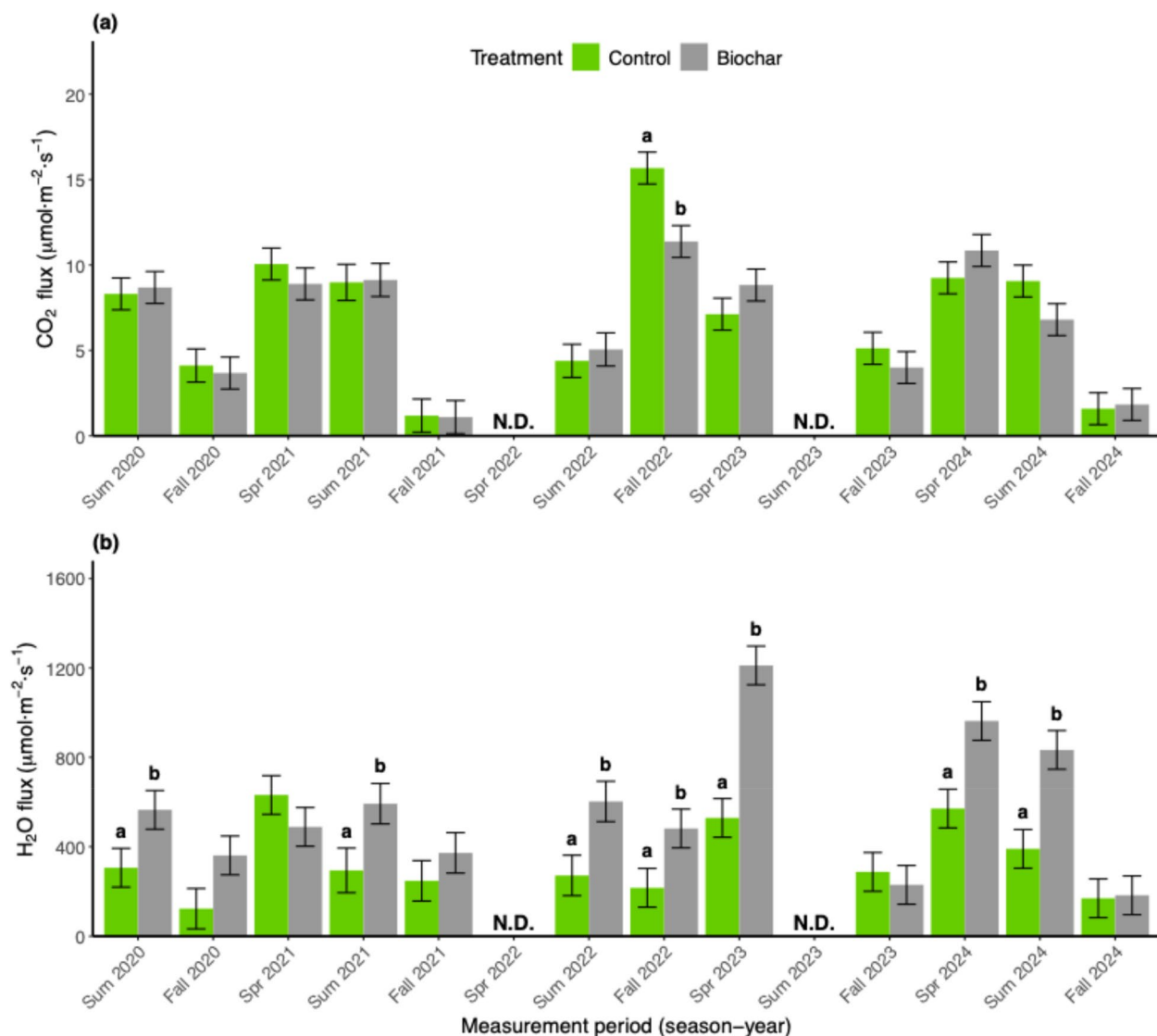


Fig. 2 Seasonal variation in mean ($\pm$ SE) (a) CO₂ and (b) H₂O fluxes from extensive green roof modules with and without biochar amendment between summer 2020 and fall 2024. Bars represent estimated marginal means derived from linear mixed-effects models with treatment and measurement period (season-year) as fixed effects and collar as a random effect. Within each measurement period, different letters indicate significant differences between biochar and control treatments based on Sidak-adjusted pairwise comparisons ($p < 0.05$). N.D. denotes periods for which measurements were not conducted (Spring 2022 and summer 2023)

3.3 Biochar increases evaporation through enhanced surface flux

Results from LMMs showed that biochar amendment had a highly significant effect on H₂O vapour flux in green roof substrates ($F_{1,255} = 29.00$, $p < 0.001$). The main effect of season was also highly significant ($F_{11,255} = 12.62$, $p < 0.001$), indicating strong seasonal variation in surface evaporation. In addition, the interaction between treatment and season was significant ($F_{11,255} = 3.63$, $p < 0.001$), indicating temporal variation in the influence of biochar on H₂O flux (Table 1).

H₂O fluxes exhibited strong seasonal and interannual variation, with consistently higher fluxes in biochar-amended modules than in those without biochar (Fig. 2b; Table S4). During the establishment years (2020–2021), H₂O fluxes were already elevated in biochar-treated modules (e.g., summer 2020: 564 ± 25 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) compared with controls (306 ± 3 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; $p = 0.05$). The largest treatment effect occurred in spring 2023, when mean fluxes in biochar modules (1211 ± 192 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) exceeded those without biochar (528 ± 63 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; $p < 0.001$) by more than twofold. Fluxes generally peaked in

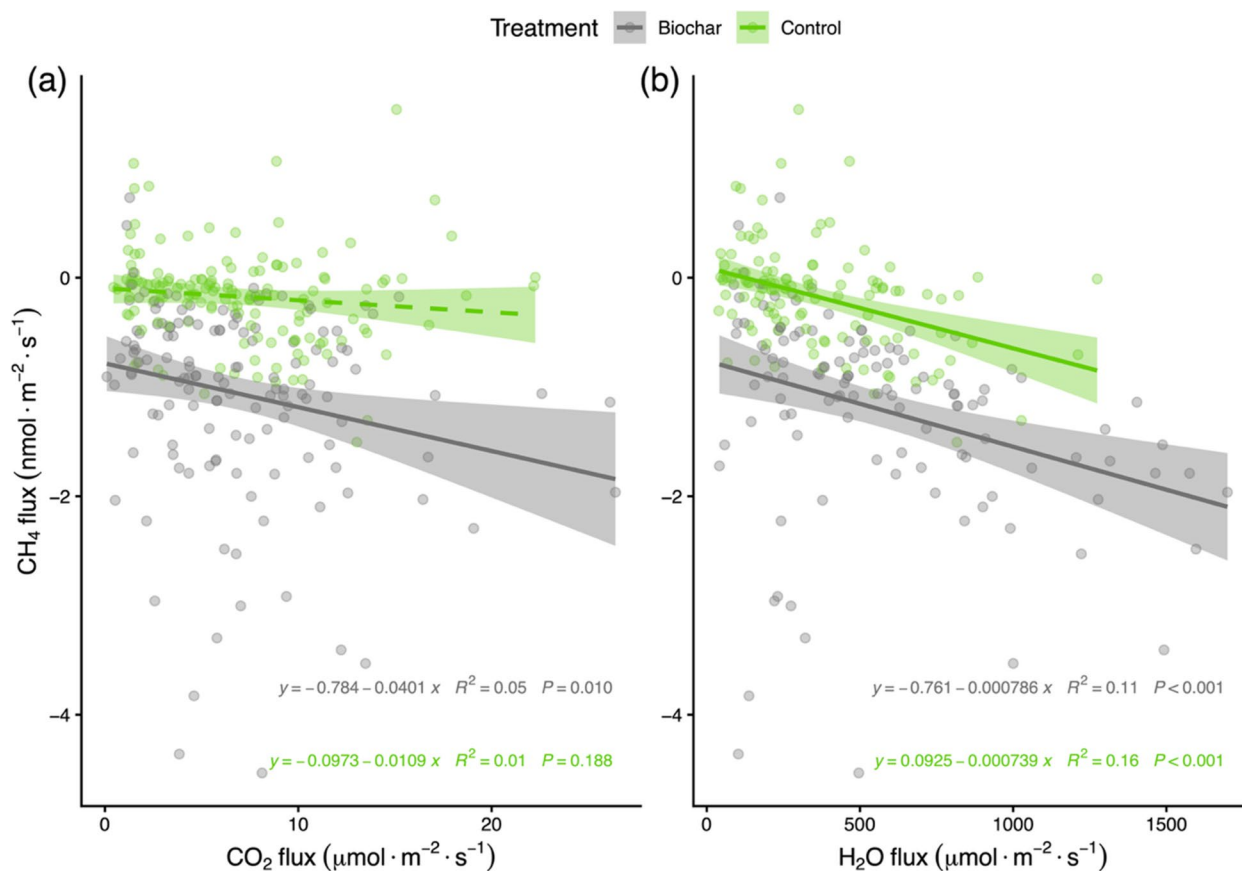


Fig. 3 Relationships between CH_4 flux and (a) CO_2 flux and (b) H_2O flux from extensive green roof substrates, based on pooled data measured across all seasons between 2020 and 2024. Regression equations, coefficients of determination (R^2), and significance levels are shown. The shaded area around the fitted regression line represents the 95% confidence interval. A solid line indicates a significant model fit, while a dashed line indicates a non-significant fit

spring and summer and declined in fall (Fig. S4). By 2024, both treatments showed reduced fluxes, though biochar-amended modules maintained higher soil evaporation ($390 \pm 178 \mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) than controls ($170 \pm 27 \mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$; $p=0.03$). Overall, biochar addition enhanced H_2O fluxes throughout the study period, especially during warm and moist seasons when evaporative potential was highest.

3.4 CH_4 uptake covaries with CO_2 and H_2O fluxes

CH_4 uptake increased significantly with both CO_2 flux ($R^2=0.05$, $p<0.01$) and H_2O flux ($R^2=0.16$, $p<0.001$) (Fig. 3). These positive relationships were stronger in biochar-amended modules, which consistently exhibited greater CH_4 uptake than those without biochar. Linear regressions between substrate moisture, temperature, EC, and pH with CH_4 flux are shown in Supplementary Fig. S5.

3.5 Integrated pathway analysis reveals controls on GHG fluxes

The piecewise structural equation model showed an acceptable fit to the data (Fisher's $C=11.17$, $df=6$, $p=0.083$) and explained a substantial proportion of the variance in CH_4 flux ($R^2=0.47$) and CO_2 flux ($R^2=0.17$), with more limited explanatory power for intermediate variables (moisture=0.11; temperature=0.03; EC=0.12; pH=0.01; H_2O flux=0.11; Fig. 4). Biochar addition exerted a strong, significant direct effect on CH_4 flux ($\beta = -0.94$), indicating enhanced CH_4 uptake under biochar-amended conditions. Biochar also significantly increased substrate moisture ($\beta=0.65$) and reduced electrical conductivity (EC; $\beta = -0.61$), and had a marginal effect on substrate pH ($\beta = -0.14$; $p<0.10$; Supplementary Fig. S6). Higher substrate moisture significantly promoted H_2O flux ($\beta=0.13$) and increased EC ($\beta=0.20$), while

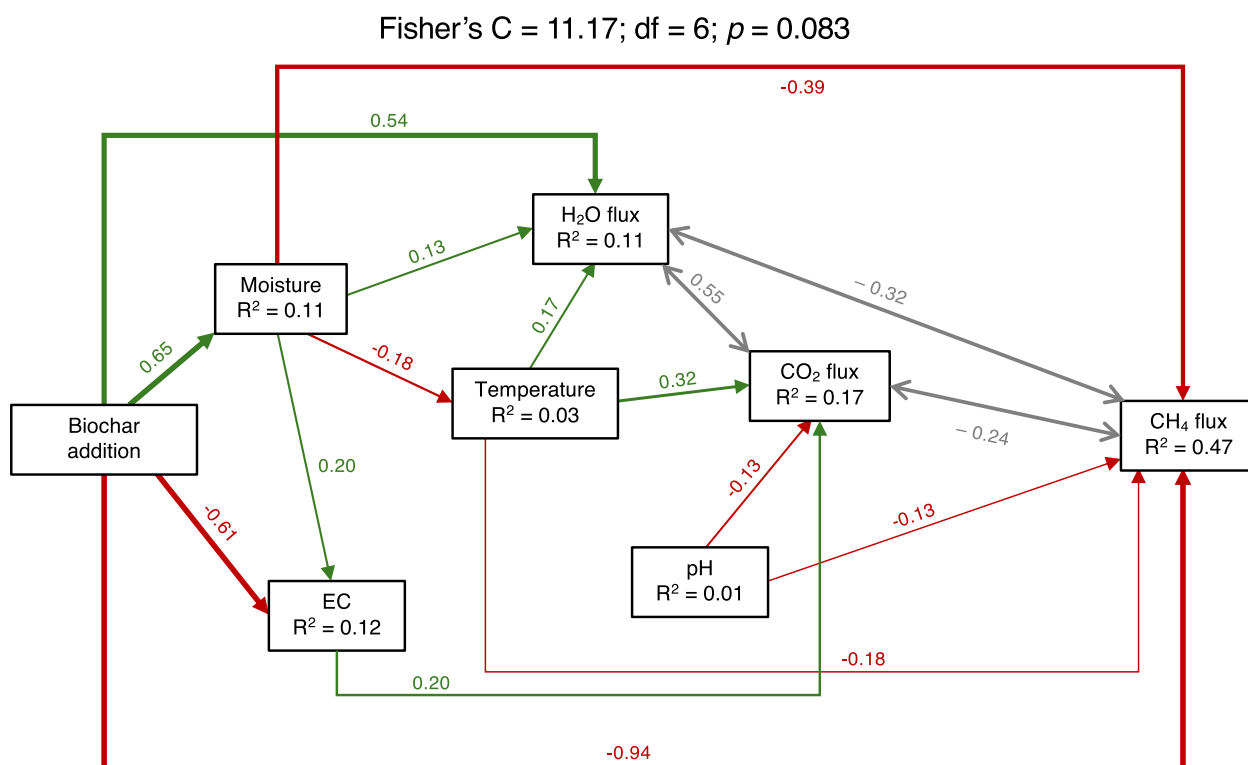


Fig. 4 Piecewise structural equation model (SEM) showing the significant pathways ($p < 0.05$) linking biochar addition, environmental drivers, and CH₄ flux from green roof substrates. Green arrows represent positive standardized path coefficients and red arrows represent negative standardized path coefficients, with arrow thickness proportional to effect size. Numbers adjacent to paths are standardized coefficients. Endogenous variables are shown with their conditional R^2 values, representing variance explained by predictors. Two-headed grey arrows denote residual correlations among gas fluxes. The non-significant p -value indicates adequate model fit (Fisher's C = 11.17, df = 6, $p = 0.083$). Non-significant causal paths are excluded here but are presented in supplementary Fig. S6. Negative standardized path coefficient for CH₄ flux indicate positive effects on CH₄ uptake

simultaneously exerting significant suppressive effects on temperature ($\beta = -0.18$). Temperature positively influenced both H₂O flux ($\beta = 0.17$) and CO₂ flux ($\beta = 0.32$) but was negatively associated with CH₄ flux ($\beta = -0.18$), indicating higher methane uptake with increased temperature. EC had a modest positive effect on CO₂ flux ($\beta = 0.20$) but was not directly linked to CH₄ flux. Substrate pH was weakly but negatively associated with both CO₂ flux ($\beta = -0.13$) and CH₄ flux ($\beta = -0.13$). However, given the marginal biochar–pH pathway and the low variance explained, pH appears to act as a significant secondary chemical correlate rather than a primary biochar-mediated control on gas exchange. Residual correlations among H₂O, CO₂, and CH₄ fluxes (grey two-headed arrows) indicate shared environmental or biological drivers not explicitly resolved as causal pathways in the model, rather than direct flux–flux regulation.

4 Discussion

4.1 Biochar promotes sustained CH₄ uptake in engineered green roof substrates

Over five years of field measurements on stonecrop-vegetated extensive green roofs, biochar amendment substantially enhanced CH₄ uptake relative to non-amended control substrates. Although the magnitude of uptake varied with season and substrate age, the direction of the biochar effect remained consistent, supporting the hypothesis that biochar strongly promotes conditions favorable for methanotrophy in organic-rich engineered media. Biochar addition also increased surface evaporation but did not elevate CO₂ emissions, indicating that enhanced CH₄ uptake was not accompanied by increased gaseous carbon losses. The sustained enhancement of CH₄ uptake across years was closely associated with higher substrate moisture and greater H₂O flux, suggesting that biochar improved gas diffusivity and maintained aerobic microsites conducive to microbial CH₄ oxidation under near-optimal moisture conditions.

The CH₄ uptake rates observed in biochar-amended substrates (up to $-1.9 \text{ nmol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) exceeded those typically reported for agricultural soils (-0.1 to $-1.0 \text{ nmol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; Le Mer and Roger 2001) and urban soils (often negligible to $-1.0 \text{ nmol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$; Groffman and Pouyat 2009), though they remained lower than growing season uptake rates reported for intact upland temperate forest soils in the region (-2.4 to $-4.2 \text{ nmol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) (Priemé and Christensen 1997; Steinkamp et al. 2001; Vantellingen et al. 2022). These comparisons highlight the potential for biochar-amended green roofs to function as meaningful CH₄ sinks within urban environments, despite their shallow profiles and engineered constraints. Only a limited number of studies have quantified GHG fluxes from green roof substrates, with most reporting minimal CH₄ uptake (Halim et al. 2022). While biochar application in urban green infrastructure has gained attention, its effect on CH₄ flux in extensive green roof systems had not been previously tested. Prior research in agricultural and urban soils largely focused on short-term experiments, where biochar generally reduced or reversed CH₄ emissions (Wu et al. 2019; Kayes et al. 2025), but evidence for long-term persistence has remained scarce. Our results provide the first five-year field demonstration that biochar can substantially strengthen the CH₄ sink capacity of green roof substrates, adding a previously unquantified climate co-benefit to urban green infrastructure.

Methane uptake generally peaked during spring and fall measurement campaigns, with the largest numerical treatment difference observed in spring 2023. Elevated uptake was also observed in spring 2024, suggesting that this pattern was not restricted to a single episode. The increasing CH₄ uptake observed during later years of the experiment may partly reflect progressive biochar aging. As biochar weathers, oxidation of particle surfaces, greater pore accessibility, and microbial colonization can enhance conditions favorable for methane-oxidizing microorganisms (Lehmann and Joseph 2015; Wang et al. 2016). Spring conditions may further promote CH₄ oxidation through moderate substrate temperatures, renewed microbial activity following winter dormancy, and moisture conditions that support both methanotrophic activity and gas diffusion (Le Mer and Roger 2001). Together, these results indicate that biochar promotes sustained methane uptake in engineered green roof substrates by stabilizing moisture availability and enhancing gas diffusivity, thereby maintaining aerobic conditions favorable for methanotrophy over time.

4.2 Biochar-mediated methane uptake pathways

Our results showed that biochar enhanced CH₄ uptake through both direct effects and significant indirect effects mediated by higher substrate moisture and H₂O flux, together creating conditions favorable for methanotrophy. Similar enhancements of CH₄ uptake following biochar application have been reported in paddy fields (Wu et al. 2019), urban soils (Kayes et al. 2025) and landfill cover materials (Chetri and Reddy 2022). These responses are commonly attributed to biochar-induced improvements in substrate physical structure, aeration, and microbial habitat. Biochar possesses a highly porous structure and extensive internal surface area that can increase air-filled porosity, improve gas diffusivity, and provide protected microsites for microbial colonization (Lehmann and Joseph 2015). In engineered green roof substrates, where methane oxidation is constrained by shallow substrate depth and rapid fluctuations in moisture and aeration, these structural modifications likely promoted conditions favorable for atmospheric CH₄ uptake.

The structural equation model revealed a significant relationship between substrate moisture and CH₄ uptake, and between biochar addition and substrate moisture, indicating that biochar improved water retention and thereby enhanced CH₄ uptake. Soil moisture strongly regulates biogeochemical processes: excessive moisture limits air-filled pore space and creates anaerobic microsites, while low moisture imposes water stress on methanotrophs (Castro et al. 1995; Borken et al. 2006). Maximum CH₄ uptake rates have been recorded in forest and grassland soils at 20–60% saturation (Bowden et al. 1998; Feng et al. 2021, 2023). While green roof substrates are unique, we found moisture generally remained within this optimal range in both treatments but was consistently higher in biochar-amended modules, indicating that biochar maintains conditions favorable for CH₄ uptake under dry conditions. We also observed a significant negative relationship between substrate temperature and CH₄ flux (Fig. 4), suggesting that higher temperatures stimulated methane oxidation. Structural equation modeling indicated that biochar exerted a positive but non-significant effect on temperature, possibly due to its low albedo increasing heat absorption, offset by greater evaporative cooling. This thermal balance resembles prior observations of biochar effects on green roof microclimates (Chen et al. 2018). We also observed non-significant relationships between CH₄ uptake and substrate EC but a significant effect of pH (Fig. 4). The average pH in biochar-treated modules (6.57 ± 0.04 to 7.53 ± 0.05 ; Table S5) was near-neutral, within the optimal range (6–8) for CH₄ oxidation (Hanson and Hanson 1996). Although biochar commonly has a liming effect on

acid soils (Zhang et al. 2025), we did not detect any effect in the present study; instead, biochar-induced pH buffering appears to have contributed to favorable conditions for methanotrophic activity.

The strong direct effect of biochar on CH₄ flux observed in the structural equation model ($\beta = -0.94$) indicates that additional mechanisms beyond the measured mediators largely accounted for enhanced CH₄ uptake. Biochar influences not only substrate moisture and aeration but also microbial habitat quality, surface chemistry, and microscale redox conditions, all of which may affect CH₄ cycling. Biochar possesses a highly porous structure and extensive internal surface area that can provide protected microsites for microbial colonization, improve oxygen diffusion, and stabilize moisture films at pore surfaces (Lehmann and Joseph 2015). In addition to its physical structure, biochar contains diverse oxygen-containing functional groups and redox-active surfaces that can influence microbial colonization, electron transfer processes, and microbial activity (Joseph et al. 2010; Lehmann and Joseph 2015). Previous studies have reported enhanced methanotroph abundance and CH₄ oxidation following biochar amendment, attributed to improved microbial habitat quality, greater aeration, increased gas diffusivity, and favorable physicochemical conditions for methane-oxidizing microorganisms (Karhu et al. 2011; Jeffery et al. 2016; Wu et al. 2019; Kayes et al. 2025). In addition, biochar may buffer rapid drying–rewetting cycles common on rooftop systems by retaining moisture while maintaining air-filled pore space, thereby sustaining favorable conditions for CH₄ oxidation. Collectively, these mechanisms provide a plausible explanation for the strong direct pathway observed between biochar addition and CH₄ uptake.

4.3 Coupling of CH₄ uptake, substrate surface evaporation, and CO₂ efflux

We found a significant relationship between CH₄ and H₂O flux (Fig. 4), with biochar also consistently enhancing H₂O flux across seasons (Fig. 2). In rooftop environments characterized by higher atmospheric evaporative demand than ground-level environments, biochar likely enhances capillary continuity and pore connectivity (Lehmann and Joseph 2009) facilitating vertical water transport in the substrate, thereby improving gas diffusivity in near-surface oxic layers and facilitating microbial CH₄ oxidation (Or et al. 2013; Smith et al. 2018). This dual effect of greater water storage coupled with improved gaseous exchange appears to sustain moisture while simultaneously enhancing oxygen diffusion, promoting long-term CH₄ uptake in biochar-amended substrates.

The observed relationships among CH₄, CO₂, and H₂O fluxes further indicate strong coupling between CH₄ uptake, evaporation, and carbon dioxide efflux in engineered green roof substrates. During Stage-I evaporation, removal of surface water increases air-filled porosity while capillary continuity maintains moisture supply to the surface (Or et al. 2013), creating oxic microsites immediately above wetter layers where CH₄ oxidation can proceed efficiently. Because methanotrophic activity is highly sensitive to oxygen availability, even modest increases in air-filled pore space can substantially enhance CH₄ uptake (Smith et al. 2018). In biochar-amended substrates, improved pore connectivity and sustained moisture likely amplified these effects, strengthening the coupling between evaporation and CH₄ uptake.

The negative relationship between CH₄ uptake and CO₂ efflux may reflect shared environmental controls on methanotrophic activity and substrate respiration. Although methanotrophs oxidize CH₄ to CO₂ (Hanson and Hanson 1996; Conrad 2007), the contribution of CH₄ oxidation to total measured CO₂ efflux was likely very small relative to overall substrate respiration. Collectively, these findings demonstrate that biochar modifies CH₄ flux not through a single control variable, but through interacting physical and biological processes that link water movement, gas diffusion, and microbial metabolism.

4.4 Biochar effects on CO₂ emissions and gaseous carbon balance in engineered green roof substrates

Despite enhancing CH₄ uptake, biochar application did not significantly increase CO₂ emissions (Fig. 2a). The tested green roof substrate was already rich in organic matter, and any transient CO₂ pulse from labile carbon likely occurred before monitoring (modules installed in 2019; measurements began 2020). CO₂ efflux showed strong seasonal patterns, higher in spring and summer, and lower in fall, with temperature exerting a significant positive effect (Osanai et al. 2015; Chen et al. 2021). Seasonal variation of CO₂ flux also reflects plant root respiration during active growth (Ben-Noah and Friedman 2018).

The absence of a sustained biochar-induced increase in CO₂ emissions suggests that enhanced CH₄ uptake did not come at the expense of greater carbon loss from the system in the form of gaseous exchange. Instead, biochar appears to shift gas-exchange dynamics toward increased CH₄ consumption while maintaining overall carbon balance. In a companion study on discharge water quality from this same experiment, biochar additions reduced discharge water turbidity and total suspended solids, suggesting that biochar additions likely

reduced C losses via leaching (Liao et al. 2024). When considered alongside the positive coupling between CH₄ uptake, evaporation, and CO₂ efflux, these results highlight the importance of integrated physical–biological controls on GHG fluxes in engineered substrates. Understanding and leveraging these interactions offers a promising pathway for designing green roof substrates that simultaneously support hydrological function, plant performance, and greenhouse gas mitigation.

4.5 Limitations and future perspectives

This study evaluated a single biochar type applied at a fixed dosage, whereas biochar properties vary widely with feedstock and pyrolysis conditions. Establishing dose–response relationships across biochar types will be essential to identify thresholds at which benefits to methane uptake plateau or unintended effects emerge. Future work should also link observed flux responses to microbial population dynamics using molecular approaches, such as functional gene assays targeting methanotroph abundance and activity, to directly resolve biological mechanisms underlying biochar-induced CH₄ uptake. Beyond these methodological gaps, our findings have clear implications for engineered substrate design. The strong coupling among biochar addition, substrate moisture, evaporation, and methane uptake suggests that biochar should be considered as a multifunctional design component, rather than a passive carbon amendment. Incorporating biochar into green roof substrates may allow designers to enhance gas diffusivity and moisture retention simultaneously, supporting methane uptake while maintaining or enhancing hydrological and plant-performance objectives. Because extensive green roofs are often managed to balance drainage and aeration under shallow substrate depths, biochar offers a practical means of stabilizing these properties over time.

While this study focused on extensive stonecrop-dominated systems, future research should evaluate biochar amendments in native plant green roofs, where greater rooting depth, rhizosphere complexity, and plant–microbe interactions may further influence CH₄ sink strength and broader ecosystem functions. Finally, our results connect biochar use in green roofs to urban wood-waste management strategies. Converting forestry and urban tree residues into biochar for engineered substrates could provide durable carbon storage while simultaneously enhancing greenhouse gas mitigation, stormwater regulation, long-term substrate performance relative to conventional organic amendments, and facilitates circular economy approaches to urban biomass management.

5 Conclusion

Pre-grown stonecrop green roofs showed low but detectable rates of CH₄ uptake; biochar amendment greatly enhanced CH₄ uptake, with effects persisting over the five-year of the study. Although uptake declined slightly in the fifth year, it remained consistently higher than in unamended substrates. Biochar-mediated enhancement of CH₄ uptake was likely driven by multiple interacting pathways, including improved substrate moisture retention, moderated moisture fluctuations, enhanced aeration and gas diffusivity, and the creation of favorable microsites for methanotrophic activity within engineered substrates. These findings demonstrate that a single biochar amendment can maintain elevated CH₄ uptake over multiple years while simultaneously improving hydrological function. Collectively, our results show that biochar-enhanced CH₄ oxidation can be integrated with the pronounced stormwater-regulation and plant-performance benefits of green roofs, supporting their role as multifunctional and climate-resilient components of urban green infrastructure.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44246-026-00296-y>.

Supplementary Material 1.

Acknowledgements

The authors gratefully acknowledge Professors Jennifer Drake and Liat Margolis, co-principal investigators on the broader research project, for their valuable leadership and logistical support. We are grateful to Malaika Mitra, Katie Monat, Jennifer Barrett, Liam Douglas, and Jovana Shrestha for their assistance in fieldwork. We thank Tony Ung for his support in maintaining the green-roof experiment. We also acknowledge Haliburton Forest and Wildlife Reserve Ltd., Gro-Bark Inc., and Bioroof Systems Inc. for providing materials and product information.

Authors' contributions

Imrul Kayes and Sean C. Thomas contributed to the conceptualization of the research. Imrul Kayes, Md Abdul Halim, Wenxi Liao, Md Rezaul Karim, and Melanie A. Sifton participated in field data collection and data curation. Imrul Kayes conducted the formal data analysis and prepared the original manuscript draft. Sean C. Thomas provided validation and supervision and contributed to project administration and funding acquisition. All authors reviewed and approved the final manuscript.

Funding

This research was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC)-CREATE and Discovery Grants awarded to Sean C. Thomas.

Data availability

The source data used to generate all graphs and charts presented in this study are publicly available via the Scholars Portal Dataverse repository at: <https://doi.org/10.5683/SP3/CANT7Q>.

Declarations

Competing interests

Sean C. Thomas is an editorial board member for Carbon Research and was not involved in the editorial review, or the decision to publish, this article. All authors declare that there are no competing interests.

Author details

¹Institute of Forestry and Conservation, John H. Daniels Faculty of Architecture, Landscape and Design, University of Toronto, 33 Willcocks Street, Toronto, ON M5S 3B3, Canada. ²School of the Environment, University of Toronto, 5 Bancroft Ave, Toronto, ON M5S 3J1, Canada. ³CredoSense Inc., B114-3600 Steeles Ave. E, Markham, ON L3R 9Z7, Canada. ⁴School of Environmental Sciences, University of Guelph, 50 Stone Road East, Guelph, ON N1G 2W1, Canada.

Received: 25 February 2026 Revised: 31 May 2026 Accepted: 12 June 2026

Published online: 20 July 2026

References

- Abeyasinghe AMSN, Lakshani MMT, Amarasinghe UDHN et al (2022) Soil-gas diffusivity-based characterization of variably saturated agricultural topsoils. *Water* 14:2900. <https://doi.org/10.3390/w14182900>
- A'saf TS, Al-Ajlouni MG, Ayad JY et al (2020) Performance of six different soilless green roof substrates for the Mediterranean region. *Sci Total Environ* 730:139182. <https://doi.org/10.1016/j.scitotenv.2020.139182>
- Bates D, Mächler M, Bolker B, Walker S (2015) Fitting linear mixed-effects models using lme4. *J Stat Softw* 67:1–48. <https://doi.org/10.18637/jss.v067.i01>
- Ben-Noah I, Friedman SP (2018) Review and evaluation of root respiration and of natural and agricultural processes of soil aeration. *Vadose Zone J* 17:170119. <https://doi.org/10.2136/vzj2017.06.0119>
- Birch HF (1958) The effect of soil drying on humus decomposition and nitrogen availability. *Plant Soil* 10:9–31. <https://doi.org/10.1007/BF01343734>
- Borken W, Savage K, Davidson EA, Trumbore SE (2006) Effects of experimental drought on soil respiration and radiocarbon efflux from a temperate forest soil. *Glob Change Biol* 12:177–193. <https://doi.org/10.1111/j.1365-2486.2005.001058.x>
- Bowden RD, Newkirk KM, Rullo GM (1998) Carbon dioxide and methane fluxes by a forest soil under laboratory-controlled moisture and temperature conditions. *Soil Biol Biochem* 30:1591–1597. [https://doi.org/10.1016/S0038-0717\(97\)00228-9](https://doi.org/10.1016/S0038-0717(97)00228-9)
- Castro MS, Steudler PA, Melillo JM et al (1995) Factors controlling atmospheric methane consumption by temperate forest soils. *Glob Biogeochem Cycles* 9:1–10. <https://doi.org/10.1029/94GB02651>
- Chen H, Ma J, Wei J et al (2018) Biochar increases plant growth and alters microbial communities via regulating the moisture and temperature of green roof substrates. *Sci Total Environ* 635:333–342. <https://doi.org/10.1016/j.scitotenv.2018.04.127>
- Chen X, Liu M, Xu Z, Wei H (2021) Influences of temperature and moisture on abiotic and biotic soil CO₂ emission from a subtropical forest. *Carbon Balance Manag* 16:18. <https://doi.org/10.1186/s13021-021-00181-8>
- Chen X, Zhu H, Bañuelos G et al (2020) Biochar reduces nitrous oxide but increases methane emissions in batch wetland mesocosms. *Chem Eng J* 392:124842. <https://doi.org/10.1016/j.cej.2020.124842>
- Chettri JK, Reddy KR (2022) Enhanced landfill methane oxidation using activated biochar. 1–10. <https://doi.org/10.1061/9780784484050.001>
- Conrad R (2007) Microbial ecology of methanogens and methanotrophs. In: *Advances in agronomy*. Academic Press, pp 1–63
- Danner D, Hagemann D, Fiedler K (2015) Mediation analysis with structural equation models: combining theory, design, and statistics. *Eur J Soc Psychol* 45:460–481. <https://doi.org/10.1002/ejsp.2106>
- De-Ville S, Menon M, Jia X, Stovin V (2018) A longitudinal microcosm study on the effects of ageing on potential green roof hydrological performance. *Water* 10:784. <https://doi.org/10.3390/w10060784>
- Ding Y, Wang M, Du X, et al (2024) The impact of elevated CO₂ on methanogen abundance and methane emissions in terrestrial ecosystems: a meta-analysis. *iScience* 27:111504. <https://doi.org/10.1016/j.isci.2024.111504>
- Feng H, Guo J, Malghani S et al (2021) Effects of soil moisture and temperature on microbial regulation of methane fluxes in a poplar plantation. *Forests* 12:407. <https://doi.org/10.3390/f12040407>
- Feng W, Wang T, Yang F et al (2023) Effects of biochar on soil evaporation and moisture content and the associated mechanisms. *Environ Sci Eur* 35:66. <https://doi.org/10.1186/s12302-023-00776-7>
- Fox J, Weisberg S, Price B (2001) car: companion to applied regression. 3.1–3
- Gale NV, Thomas SC (2019) Dose-dependence of growth and ecophysiological responses of plants to biochar. *Sci Total Environ* 658:1344–1354. <https://doi.org/10.1016/j.scitotenv.2018.12.239>
- Groffman PM, Pouyat RV (2009) Methane uptake in urban forests and lawns. *Environ Sci Technol* 43:5229–5235. <https://doi.org/10.1021/es803720h>
- Gunzler D, Chen T, Wu P, Zhang H (2013) Introduction to mediation analysis with structural equation modeling. *Shanghai Arch Psychiatry* 25:390–394. <https://doi.org/10.3969/j.issn.1002-0829.2013.06.009>
- Guo F, Zhang J, Yang X et al (2020) Impact of biochar on greenhouse gas emissions from constructed wetlands under various influent chemical oxygen demand to nitrogen ratios. *Bioresour Technol* 303:122908. <https://doi.org/10.1016/j.biortech.2020.122908>
- Halim MA, Bieser JMH, Thomas SC (2024) Large, sustained soil CO₂ efflux but rapid recovery of CH₄ oxidation in post-harvest and post-fire stands in a mixedwood boreal forest. *Sci Total Environ* 930:172666. <https://doi.org/10.1016/j.scitotenv.2024.172666>
- Halim MA, Vantellingen J, Gorgolewski AS et al (2022) Greenhouse gases and green roofs: carbon dioxide and methane fluxes in relation to substrate characteristics. *Urban Ecosyst* 25:487–498. <https://doi.org/10.1007/s11252-021-01166-8>
- Hanson RS, Hanson TE (1996) Methanotrophic bacteria. *Microbiol Rev* 60:439–471. <https://doi.org/10.1128/mr.60.2.439-471.1996>
- Hill J, Drake J, Sleep B, Margolis L (2017) Influences of four extensive green roof design variables on stormwater hydrology. *J Hydrol Eng* 22:04017019. [https://doi.org/10.1061/\(ASCE\)HE.1943-5584.0001534](https://doi.org/10.1061/(ASCE)HE.1943-5584.0001534)
- Hoffmann M, Schulz-Hanke M, Garcia Alba J et al (2017) A simple calculation algorithm to separate high-resolution CH₄ flux measurements into ebullition- and diffusion-derived components. *Atmospheric Meas Tech* 10:109–118. <https://doi.org/10.5194/amt-10-109-2017>
- Hüppi R, Felber R, Krauss M et al (2018) Restricting the nonlinearity parameter in soil greenhouse gas flux calculation for more reliable flux estimates. *PLoS ONE* 13:e0200876. <https://doi.org/10.1371/journal.pone.0200876>
- Jeffery S, Verheijen FGA, Kammann C, Abalos D (2016) Biochar effects on methane emissions from soils: a meta-analysis. *Soil Biol Biochem* 101:251–258. <https://doi.org/10.1016/j.soilbio.2016.07.021>
- Joseph SD, Camps-Arbestain M, Lin Y et al (2010) An investigation into the reactions of biochar in soil. *Soil Res* 48:501–515. <https://doi.org/10.1071/SR10009>
- Jovani-Sancho AJ, Cummins T, Byrne KA (2018) Soil respiration partitioning in afforested temperate peatlands. *Biogeochemistry* 141:1–21. <https://doi.org/10.1007/s10533-018-0496-0>
- Karhu K, Mattila T, Bergström I, Regina K (2011) Biochar addition to agricultural soil increased CH₄ uptake and water holding capacity – results from a short-term pilot field study. *Agric Ecosyst Environ* 140:309–313. <https://doi.org/10.1016/j.agee.2010.12.005>
- Kayes I, Halim MA, Thomas SC (2025) Biochar mitigates methane emissions from organic mulching in urban soils: evidence from a long-term mesocosm experiment. *J Environ Manage* 376:124525. <https://doi.org/10.1016/j.jenvman.2025.124525>
- Krulwich TA, Hicks DB, Swartz T, Ito M (2007) Bioenergetic adaptations that support alkaliphily. In: *Physiology and biochemistry of extremophiles*. John Wiley & Sons, Ltd, pp 311–329
- Le Mer J, Roger P (2001) Production, oxidation, emission and consumption of methane by soils: a review. *Eur J Soil Biol* 37:25–50. [https://doi.org/10.1016/S1164-5563\(01\)01067-6](https://doi.org/10.1016/S1164-5563(01)01067-6)
- Lee J, Kwon EE (2024) Biochar in green roofs. *J Build Eng* 89:109272. <https://doi.org/10.1016/j.jobe.2024.109272>
- Lefcheck JS (2016) piecewiseSEM: Piecewise structural equation modeling in R for ecology, evolution, and systematics. *Methods Ecol Evol* 7:573–579. <https://doi.org/10.1111/2041-210X.12512>
- Lehmann J, Joseph S (eds) (2009) *Biochar for environmental management: science and technology*. Earthscan, London ; Sterling, VA

- Lehmann J, Joseph S (eds) (2015) Biochar for environmental management: science, technology and implementation, Second edition. Routledge, Taylor & Francis Group, London ; New York
- Lehmann P, Merlin O, Gentile P, Or D (2018) Soil Texture effects on surface resistance to bare-soil evaporation. *Geophys Res Lett* 45:10,398–10,405. <https://doi.org/10.1029/2018GL078803>
- Li H, Lin L, Peng Y et al (2024) Biochar's dual role in greenhouse gas emissions: nitrogen fertilization dependency and mitigation potential. *Sci Total Environ* 917:170293. <https://doi.org/10.1016/j.scitotenv.2024.170293>
- Liao W, Halim MA, Kayes I et al (2023) Biochar benefits green infrastructure: global meta-analysis and synthesis. *Environ Sci Technol* 57:15475–15486. <https://doi.org/10.1021/acs.est.3c04185>
- Liao W, Sidhu V, Sifton MA et al (2024) Biochar and vegetation effects on discharge water quality from organic-substrate green roofs. *Sci Total Environ* 922:171302. <https://doi.org/10.1016/j.scitotenv.2024.171302>
- Liao W, Sifton MA, Thomas SC (2022) Biochar granulation reduces substrate erosion on green roofs. *Biochar* 4:61. <https://doi.org/10.1007/s42773-022-00186-7>
- LI-COR (2015) Using the LI-8100A soil gas flux system and the LI-8150 multiplexer. LI-COR Biosciences, Nebraska, U.S.
- Mar KA, Unger C, Walderdorff L, Butler T (2022) Beyond CO₂ equivalence: the impacts of methane on climate, ecosystems, and health. *Environ Sci Policy* 134:127–136. <https://doi.org/10.1016/j.envsci.2022.03.027>
- Mihalakakou G, Souliotis M, Papadaki M et al (2023) Green roofs as a nature-based solution for improving urban sustainability: progress and perspectives. *Renew Sustain Energy Rev* 180:113306. <https://doi.org/10.1016/j.rser.2023.113306>
- Nan Q, Wang C, Wang H et al (2020) Mitigating methane emission via annual biochar amendment pyrolyzed with rice straw from the same paddy field. *Sci Total Environ* 746:141351. <https://doi.org/10.1016/j.scitotenv.2020.141351>
- United Nations (2025) 68% of the world population projected to live in urban areas by 2050, says UN. In: U. N. <https://www.un.org/uk/desa/68-world-population-projected-live-urban-areas-2050-says-un>. Accessed 16 Sep 2025
- Oberndorfer E, Lundholm J, Bass B et al (2007) Green roofs as urban ecosystems: ecological structures, functions, and services. *Bioscience* 57:823–833. <https://doi.org/10.1641/B571005>
- Or D, Lehmann P, Shahraeeni E, Shokri N (2013) Advances in soil evaporation physics—a review. *Vadose Zone J* 12(vzj2012):0163. <https://doi.org/10.2136/vzj2012.0163>
- Osanaï Y, Janes JK, Newton PCD, Hovenden MJ (2015) Warming and elevated CO₂ combine to increase microbial mineralisation of soil organic matter. *Soil Biol Biochem* 85:110–118. <https://doi.org/10.1016/j.soilbio.2015.02.032>
- Priemé A, Christensen S (1997) Seasonal and spatial variation of methane oxidation in a Danish spruce forest. *Soil Biol Biochem* 29:1165–1172. [https://doi.org/10.1016/S0038-0717\(97\)00038-2](https://doi.org/10.1016/S0038-0717(97)00038-2)
- R Core Team (2024) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://cran.r-project.org/doc/manuals/r-release/fullrefman.pdf>
- Saade J, Cazares SP, Liao W, et al (2023) Influence of biochar amendment on runoff retention and vegetation cover for extensive green roofs. In: Proceedings of the Canadian Society of Civil Engineering Annual Conference 2022. Springer, Cham, pp 1117–1132
- Saunio M, Stavert AR, Poulter B et al (2020) The global methane budget 2000–2017. *Earth Syst Sci Data* 12:1561–1623. <https://doi.org/10.5194/essd-12-1561-2020>
- Schurman JS, Thomas SC (2021) Linking soil CO₂ efflux to individual trees: size-dependent variation and the importance of the birch effect. *Soil Syst* 5:7. <https://doi.org/10.3390/soilsystems5010007>
- Shafique M, Xue X, Luo X (2020) An overview of carbon sequestration of green roofs in urban areas. *Urban For Urban Green* 47. <https://doi.org/10.1016/j.ufug.2019.126515>
- Smith KA, Ball T, Conen F et al (2018) Exchange of greenhouse gases between soil and atmosphere: interactions of soil physical factors and biological processes. *Eur J Soil Sci* 69:10–20. <https://doi.org/10.1111/ejss.12539>
- Spokas KA (2010) Review of the stability of biochar in soils: predictability of O:C molar ratios. *Carbon Manag* 1:289–303. <https://doi.org/10.4155/cmt.10.32>
- Steinkamp R, Butterbach-Bahl K, Papen H (2001) Methane oxidation by soils of an N limited and N fertilized spruce forest in the Black Forest, Germany. *Soil Biol Biochem* 33:145–153. [https://doi.org/10.1016/S0038-0717\(00\)00124-3](https://doi.org/10.1016/S0038-0717(00)00124-3)
- Teemusk A, Kull A, Kanal A, Mander Ü (2019) Environmental factors affecting greenhouse gas fluxes of green roofs in temperate zone. *Sci Total Environ* 694:133699. <https://doi.org/10.1016/j.scitotenv.2019.133699>
- Trotsenko YA, Khmelenina VN (2002) Biology of extremophilic and extremotolerant methanotrophs. *Arch Microbiol* 177:123–131. <https://doi.org/10.1007/s00203-001-0368-0>
- Vantellingen J, Gorgolewski AS, Thomas SC (2022) Post-harvest recovery of soil methane oxidation on skid trails and landings in a managed northern hardwood forest. *For Ecol Manage* 515:120202. <https://doi.org/10.1016/j.foreco.2022.120202>
- Wagner-Riddle C, Congreves KA, Abalos D et al (2017) Globally important nitrous oxide emissions from croplands induced by freeze–thaw cycles. *Nat Geosci* 10:279–283. <https://doi.org/10.1038/ngeo2907>
- Wang J, Xiong Z, Kuzyakov Y (2016) Biochar stability in soil: meta-analysis of decomposition and priming effects. *GCB Bioenergy* 8:512–523. <https://doi.org/10.1111/gcbb.12266>
- Welles, J. M., Demetriades-Shah, T. H., & McDermitt, D. K. (2001) Considerations for measuring ground CO₂ effluxes with chambers. *Chem Geol* 177(1–2), 3–13. [https://doi.org/10.1016/S0009-2541\(00\)00388-0](https://doi.org/10.1016/S0009-2541(00)00388-0)
- Whittinghill LJ, Rowe DB, Schutzki R, Cregg BM (2014) Quantifying carbon sequestration of various green roof and ornamental landscape systems. *Landsc Urban Plan* 123:41–48. <https://doi.org/10.1016/j.landurbplan.2013.11.015>
- Wooster EIF, Fleck R, Torpy F et al (2022) Urban green roofs promote metropolitan biodiversity: a comparative case study. *Build Environ* 207:108458. <https://doi.org/10.1016/j.buildenv.2021.108458>
- Wu Z, Song Y, Shen H et al (2019) Biochar can mitigate methane emissions by improving methanotrophs for prolonged period in fertilized paddy soils. *Environ Pollut* 253:1038–1046. <https://doi.org/10.1016/j.envpol.2019.07.073>
- Xue M, Farrell C (2020) Use of organic wastes to create lightweight green roof substrates with increased plant-available water. *Urban For Urban Green* 48:126569. <https://doi.org/10.1016/j.ufug.2019.126569>
- Yang Y, Davidson CI (2021) Green roof aging effect on physical properties and hydrologic performance. *J Sustain Water Built Environ* 7:04021007. <https://doi.org/10.1061/JSWBAY.0000949>
- Yao X, Wang J, Hu B (2023) How methanotrophs respond to pH: a review of ecophysiology. *Front Microbiol* 13:1034164. <https://doi.org/10.3389/fmicb.2022.1034164>
- Zhang B, Tang L, Chen Z, Chen X, You L, Pan R, Chen T, Liu Y, Lin W, and Huang, J. (2025) Comparative and synergistic impacts of lime and biochar on soil properties, nitrogen transformation, and microbial function in acidic soils under tobacco cropping. *Front Plant Sci*. 16:1530128. <https://doi.org/10.3389/fpls.2025.1530128>
- Zheng J, Roy Chowdhury T, Yang Z et al (2018) Impacts of temperature and soil characteristics on methane production and oxidation in Arctic tundra. *Biogeosciences* 15:6621–6635. <https://doi.org/10.5194/bg-15-6621-2018>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.